Assignment 5

Write a C program that creates **three processes** using fork() and establishes communication between them using pipes.The processes shall perform the following tasks:

Process 1: File Reader

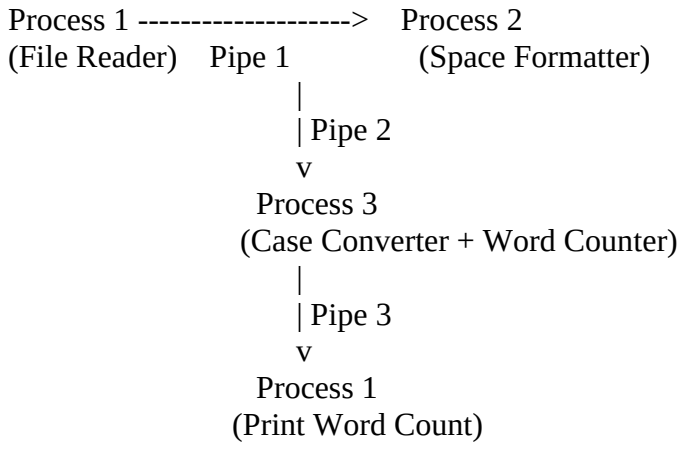
- Ask the user to enter the name/path of an input file.
- Open and read the contents of the file.
- Send the file contents to **Process 2** through a pipe.
- After receiving the word count from **Process 3**, print the total number of words.

Process 2 :Text Formatter

Process 2 shall receive the text from Process 1 through a pipe and process it according to the following rules:

- Remove unnecessary/multiple spaces.
- There should be **only one blank space between two words**.
- There should be **no blank space before punctuation marks**
- There should be **exactly one blank space after a punctuation mark**, wherever another word/character follows.
- Send the formatted text to **Process 3** through another pipe.

Process 3 : Case Converter and Word Counter: Process 3 shall:

- Receive the formatted text from Process 2.
- Convert the **first character of every word to uppercase**.
- Convert all remaining characters of each word to **lowercase**.
- Preserve punctuation marks appropriately.
- Print the final formatted text.
- Count the total number of words.
- Send the word count back to **Process 1** through a pipe

The communication should follow this structure: